Gajjarapu Satyanarayana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.50
Tirumala Junior College, Katheru	Class XII (Board of Intermediate Education, Andhra Pradesh)	2024	97.60%
The Future Kid's School, Rajahmundry	Class X (Indian Certificate of Secondary Education, ICSE)	2022	93.60%

SCHOLASTIC ACHIEVEMENTS

Received the **Best Student Award** for the *Academic Year 2017-18*.

Won a Gold Medal in the **SOF International Mathematics Olympiad (IMO)** at the school level and secured *Zonal Rank 377* in Andhra Pradesh & Telangana.

POSITIONS OF RESPONSIBILITY

Class Representative (CR), B.Tech in Artificial Intelligence 2024-2028 Batch.

Teaching Assistant for the course *Matrix Theory* under *Dr. GVV Sharma*.

- Helped students with solving problems, conducted vivas and corrected assignments.

Undergraduate Buddy, *Sunshine*, the Counselling Cell of IIT Hyderabad.

Undergraduate Mentor, *Sunshine*, the Counselling Cell of IIT Hyderabad.

SKILLS

Programming Languages: Python, C, C++, HTML/CSS, SQL.

Frameworks: PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas, Matplotlib, OpenCV.

Technologies: GitHub/Git, LATEX, PostgreSQL, MySQL, MongoDB.

RELEVANT COURSES

Artificial Intelligence: Foundations of Machine Learning, Deep Learning, Optimization - I & II, Concentration Inequalities, Programming for AI, AI for Humanity.

Computer Science: Data Structures & Applications, Algorithms, Discrete Mathematics for Computer Science, Database Management Systems - I.

Mathematics: Numerical Analysis, Probability and Random Variables, Differential Equations, Introduction to Statistics, Series of Functions, Vector Calculus, Introduction to Metric Spaces, Calculus - I & II.

Electrical: Matrix Theory, Linear Systems & Signal Processing, Control Systems.

PROJECTS

Voice & Gamepad - Controlled Unmanned Ground Vehicle (UGV) under Prof. *Dr. GVV Sharma* (Link)

- Built a low-cost prototype UGV to explore hardware platforms for porting AI and robotics control algorithms.
- Used an ESP32 microcontroller with an L293D motor driver IC to control DC drive motors and wireless communication.
- Programmed in C++ via PlatformIO to receive Bluetooth commands and integrated Android app interface (Dabble and Arduino Bluetooth Controller) supporting gamepad and voice-command modes (forward, back, left, right, stop).
- Assembled the complete hardware stack end-to-end - chassis, motor mounts, wiring, and power supply - delivering a functional, low-cost robotic prototype.

AI Chatbot - Hugging Face Inference API + Gradio (Link)

- Built a conversational AI chatbot using the Hugging Face Inference API to query a hosted LLM and served it through an interactive Gradio web interface.
- Implemented multi-turn conversation handling by maintaining a persistent message history (system, user, and assistant roles) across chat turns for context-aware responses.

Fair Bayesian Active Learning | *AI for Humanity Course Project* (Link)

- Designed three fairness-aware active learning strategies (-Aggregate, Nested, Nested-Append) combining posterior entropy and demographic parity in a Bayesian logistic regression framework.
- Built a lightweight Metric Fair Active Learning (MFAL) engine on the Give Me Some Credit dataset.
- Built the full experimental pipeline in Python (scikit-learn, fairlearn) with visualizing fairness and performance metrics.

Mini-ML Library & Autograd Engine from scratch | *Foundations of Machine Learning Course Project* (Link)

- Architected a modular ML library from scratch with clean submodules following the scikit-learn convention.
- Extended the library with a custom autograd engine and nn subpackage (Linear, ReLU, Sigmoid, SGD, CrossEntropyLoss) for neural network support.
- Validated the library by benchmarking 5 model types (OvR Logistic, Softmax variants, MLP) on Fashion-MNIST.

EXTRACURRICULAR

- Member of the **National Sports Organization (NSO) -** Cricket, IIT Hyderabad.
- Have taken part in **Cricket** team in various competitions including *Institute Cricket League (ICL)* and *Diesta*, Inter-Departmental Cultural & Sports Fest of IIT Hyderabad.
- Secured 2nd place in the Class X Intra Murals Volleyball Competition.
- Completed **Prathamika** and **Madhyamika** examinations conducted by the Dakshina Bharat Hindi Prachar Sabha.